

ALIGUDARZ, Iran, Islamic Republic of

WMO: 407830

Lat: 33.4080N

Lon: 49.7032E

Elev: 6634

StdP: 11.50

Time Zone: 3.50 (IRN)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.2	(c) 15.4	(d) -2.9	(e) 6.0	(f) 26.3	(g) 1.3	(h) 7.5	(i) 32.2	(j) 33.1	(k) 36.4	(l) 27.8	(m) 38.8	(n) 12.3	(o) 120	(p) 0.593

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 25.1	(c) 94.2	(d) 58.4	(e) 92.3	(f) 57.8	(g) 90.5	(h) 57.0	(i) 60.2	(j) 89.9	(k) 59.2	(l) 88.8	(m) 58.2	(n) 87.4	(o) 11.4	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
48.5	64.7	66.7	46.4	59.8	65.6	44.7	56.0	64.0	29.7	90.1	28.9	88.7	28.1	88.0	68.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years					
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.1	22.9	20.3	4.8	99.6	6.6	3.1	0.1	101.8	-3.8	103.6	-7.5	105.4	-12.3	107.6	
			DB	4.4	63.0	7.1	2.3	-0.7	64.6	-4.9	65.9	-8.9	67.2	-14.0	68.8
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.4	31.0	36.1	44.8	51.8	61.7	72.2	78.3	77.3	69.5	58.3	45.0	37.1
	DBStd	17.03	8.60	7.37	6.79	5.91	4.69	4.27	3.71	3.28	3.88	5.18	6.06	7.14
	HDD50	1790	590	390	187	47	0	0	0	0	0	4	171	401
	HDD65	4699	1055	809	625	396	123	3	0	0	8	214	601	865
	CDD50	3750	0	1	27	102	362	667	879	847	584	260	20	1
	CDD65	1182	0	0	0	0	20	219	414	382	142	5	0	0
	CDH74	14977	0	0	0	9	276	2821	5228	4743	1817	83	0	0
	CDH80	6574	0	0	0	0	22	1126	2625	2268	532	1	0	0
(14) Wind	WSAvg	10.0	9.6	9.7	10.3	9.8	10.0	10.1	10.8	11.0	9.9	9.8	9.0	9.5
(15) Precipitation	PrecAvg	17.0	2.2	2.1	2.7	3.0	1.1	0.1	0.1	0.0	0.1	0.9	1.8	2.8
	PrecMax	24.0	4.6	6.2	5.9	7.0	4.7	0.6	1.2	0.2	0.9	3.6	8.1	6.2
	PrecMin	10.9	0.7	0.5	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.7
	PrecStd	3.6	0.9	1.3	1.6	1.8	1.0	0.2	0.2	0.0	0.2	1.1	1.5	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.0	57.8	69.2	75.2	82.7	93.9	97.1	95.7	89.2	78.9	66.6	59.1
		MCWB	37.9	40.9	45.7	48.7	54.1	58.7	59.8	58.7	55.3	51.9	45.7	41.3
	2%	DB	48.8	53.4	64.8	69.7	79.7	91.4	94.9	93.5	87.2	76.4	62.6	54.6
		MCWB	37.0	38.4	43.5	49.1	53.1	57.4	58.9	57.8	54.9	50.5	44.0	39.7
	5%	DB	45.8	50.5	60.7	66.6	77.2	88.9	93.1	91.7	85.5	74.0	59.5	51.2
		MCWB	35.6	37.3	42.0	47.5	52.1	56.2	58.2	57.3	54.3	49.5	43.2	38.4
	10%	DB	43.1	47.3	57.3	63.8	74.6	86.3	91.2	89.9	83.3	71.3	56.2	48.1
		MCWB	34.8	36.6	40.7	46.2	51.6	55.2	57.6	56.5	53.5	48.4	41.8	37.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.0	43.2	47.2	52.2	57.2	61.2	62.0	60.9	57.9	53.7	48.5	44.2
		MCDB	47.2	51.9	62.4	68.5	77.2	86.3	92.0	90.8	85.4	73.8	59.1	53.0
	2%	WB	39.0	41.0	45.0	50.4	55.2	59.2	60.6	59.8	56.4	52.0	46.6	41.8
		MCDB	45.3	50.4	58.4	65.5	75.1	87.0	91.1	89.7	83.6	72.4	57.9	50.6
	5%	WB	37.4	39.0	43.5	48.9	53.9	57.7	59.4	58.7	55.2	50.7	44.9	40.1
		MCDB	43.3	47.5	56.6	63.2	72.9	85.5	89.7	88.6	82.7	70.4	55.4	48.5
	10%	WB	35.7	37.2	42.3	47.6	52.7	56.5	58.4	57.7	54.1	49.4	43.4	38.3
		MCDB	42.5	45.7	54.8	60.8	70.8	84.0	88.3	87.4	81.2	67.9	53.4	46.0
(35) Mean Daily Temperature Range	MDBR		16.0	16.7	18.6	19.1	21.9	26.7	25.1	25.5	27.4	22.8	18.2	17.0
	5% DB	MCDBR	18.5	20.9	22.8	22.5	25.9	29.4	27.2	27.5	28.7	26.1	23.0	21.3
		MCWBR	11.0	11.2	10.3	9.3	10.3	11.9	10.8	11.2	12.8	12.7	11.4	11.7
	5% WB	MCDBR	14.9	17.2	18.4	20.0	23.9	27.5	25.3	25.3	27.3	23.5	18.0	17.8
		MCWBR	10.2	10.5	9.7	9.1	9.9	11.9	10.5	10.8	12.6	11.6	9.4	10.5
(40) Clear-Sky Solar Irradiance	taub		0.274	0.317	0.378	0.472	0.461	0.435	0.482	0.438	0.368	0.379	0.319	0.275
	taud		2.426	2.297	2.097	1.872	1.925	2.001	1.892	2.003	2.190	2.132	2.334	2.464
	Ebn at Noon		301	296	283	260	265	270	257	267	283	268	279	295
	Edh at Noon		29	36	48	63	61	56	62	55	43	43	31	26
(44) All-Sky Solar Radiation	RadAvg		950	1231	1610	1853	2219	2632	2458	2319	2043	1504	1046	853
(45)	RadStd		73	117	118	156	158	99	112	91	77	101	95	99

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	+1.30	-1.53	-2.18	N/A	N/A	N/A	N/A